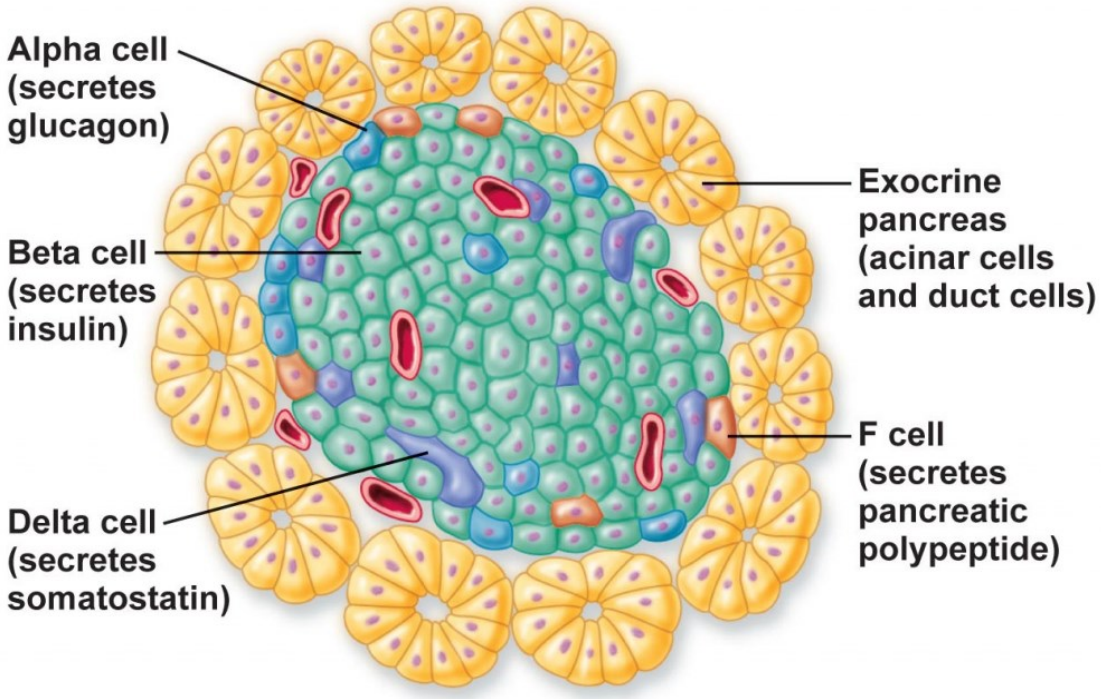
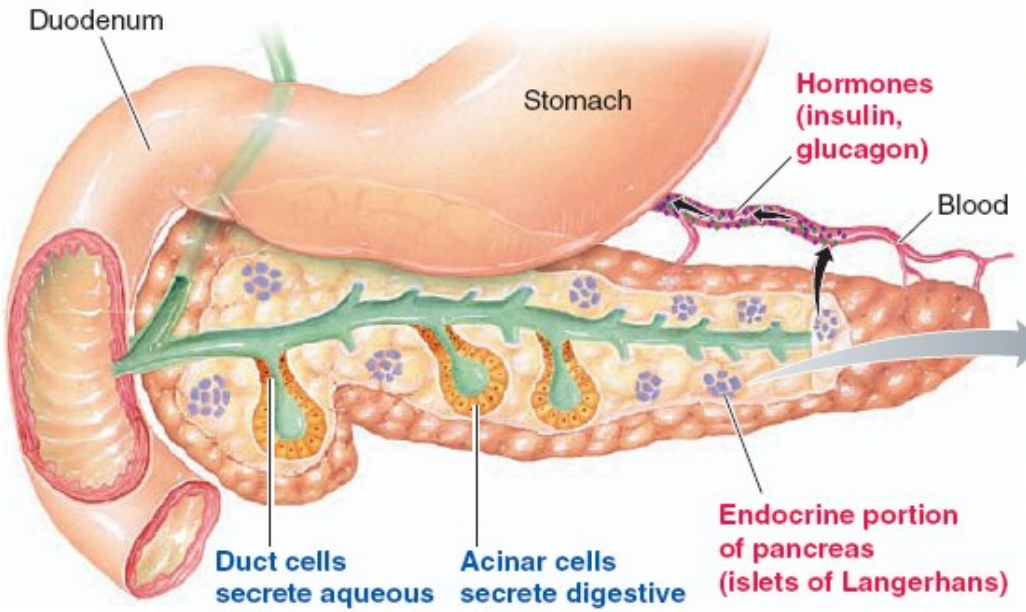
Hormone of Pancreatic islets

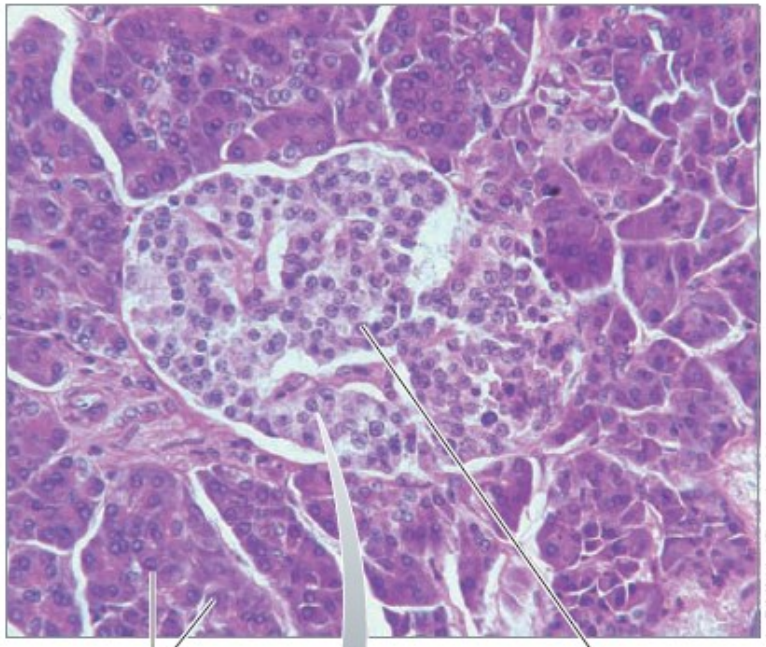


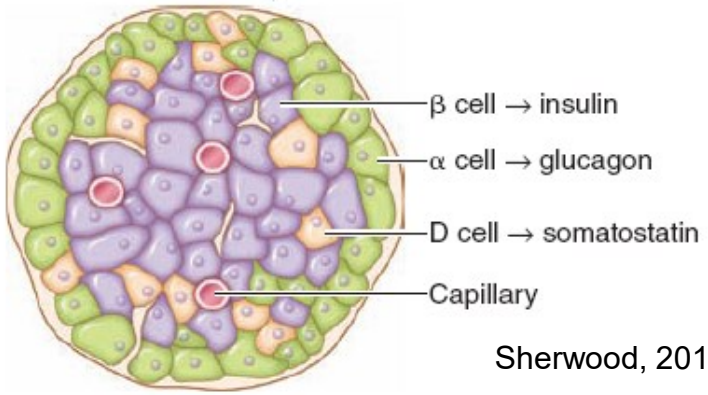
Assoc. Prof. Pissared Muangnil, Ph.D.
Faculty of Science, PSU

Pancreas



(a) Location and structure of pancreas





(b) Cell types in islet of Langerhans

- Humans have 1–2 million islets. (1–2% of total pancreatic mass)
- Blood from the islets drains into the hepatic portal vein.

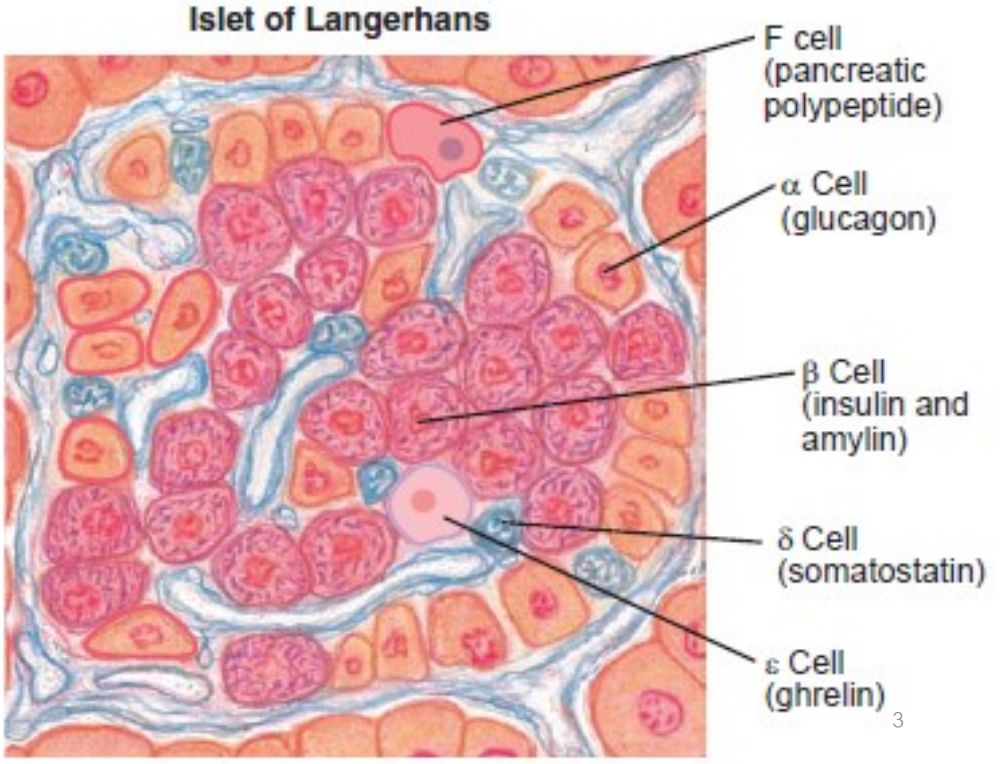
Sherwood, 2012

● FIGURE 17-24 Location and structure of the pancreas and cell types in the islets of Langerhans.

Major cell types in a typical islet of Langerhans

Islets of Langerhans = functional units of the endocrine pancreas

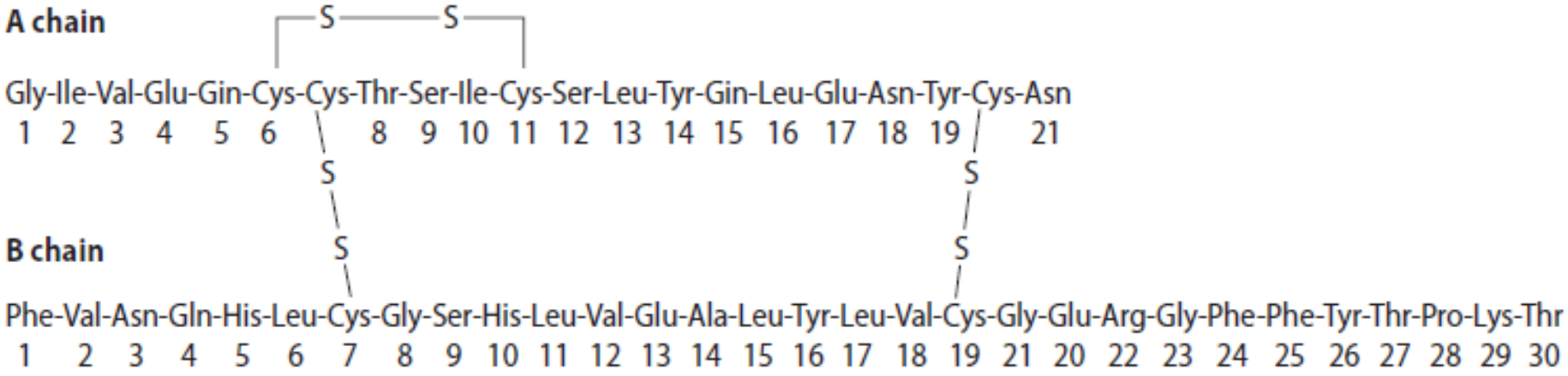
1. A or **α cell** (10%) secretes **glucagon.**
2. B or **β cell** (3/4) secretes **insulin**, proinsulin, C-peptide & amylin.
3. D or **δ cell** (5%) secretes **somatostatin.**
4. **F cell** secretes **pancreatic polypeptide** (PP).
5. **ε cell** secretes **ghrelin.**



Rhoades and Bell, 2013

Insulin

- A polypeptide containing 2 chains of amino acids linked by disulfide bridges

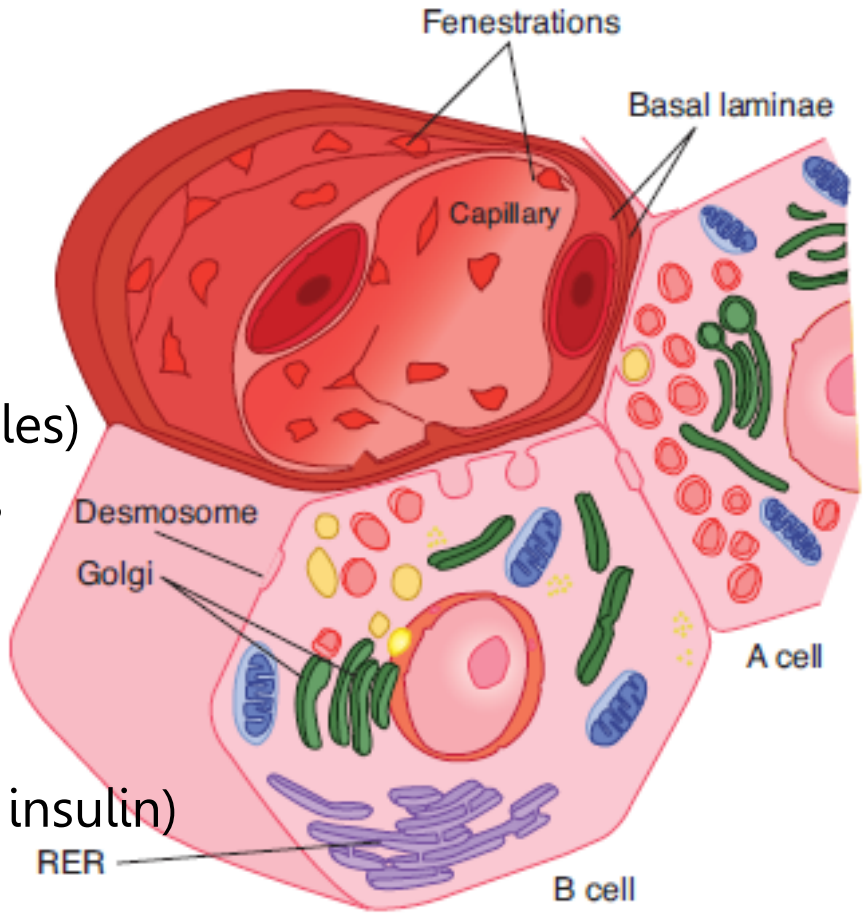


Barrett et al., 2010

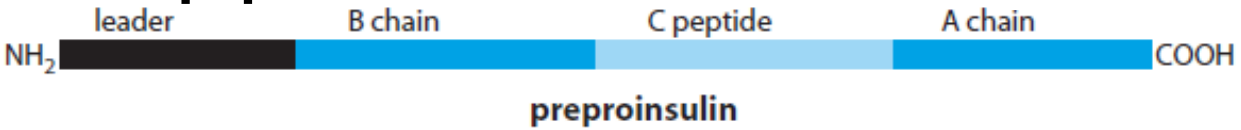
- Half-life is ~5 min.
- Destroyed by proteases (insulinase) in the endosomes formed by the endocytotic process in the **liver**, kidney & other tissues.

Insulin biosynthesis & secretion

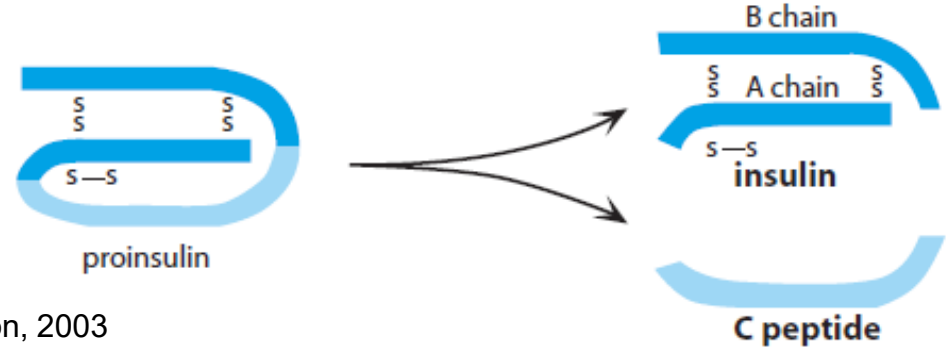
- Synthesized in RER of B cells
- Transported to Golgi apparatus (packaged into membrane-bound granules)
- Move to plasma membrane by a process involving microtubules, expelled by exocytosis
- Cross basal lamina of B cell & capillary
- **Secretion of insulin, proinsulin** (5% of insulin) **& C peptide**



Barrett et al., 2010



preproinsulin

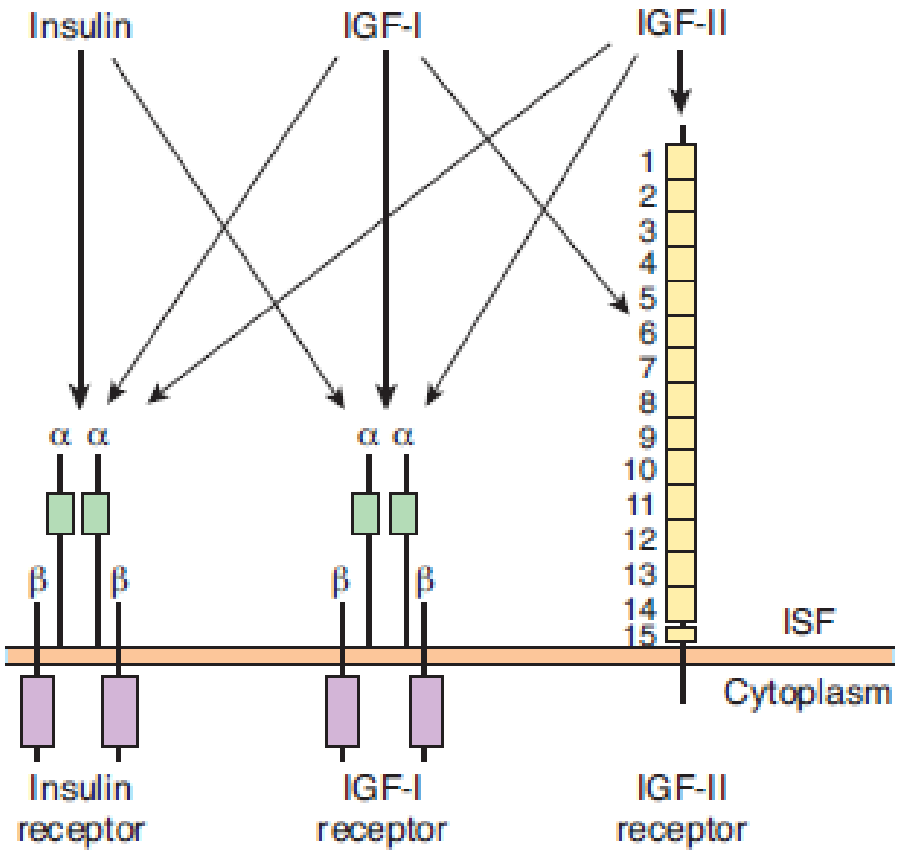


Johnson, 2003

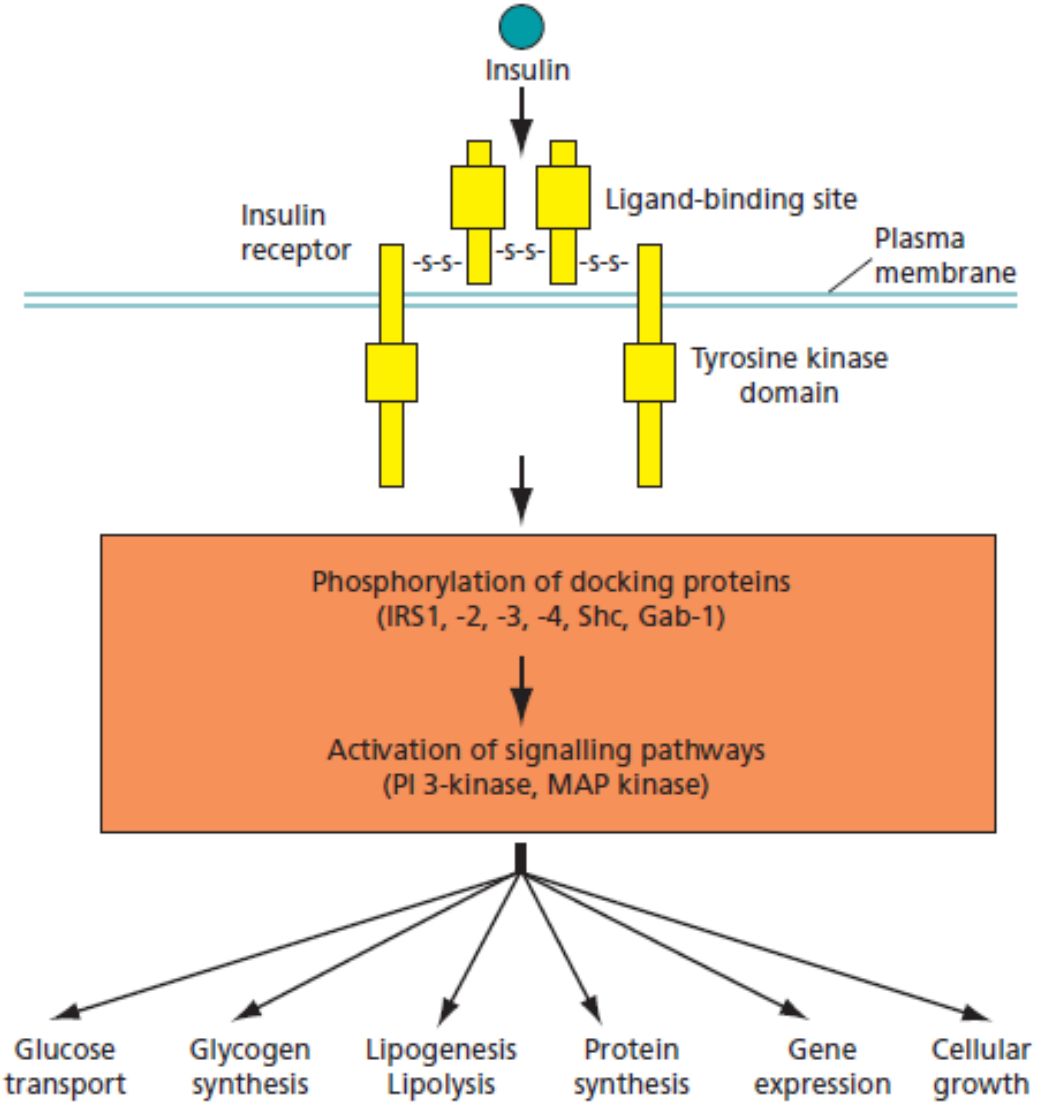
- Proinsulin: modest insulin-like activity
- C peptide: no established biological action, but useful marker for insulin secretion

Insulin receptors

- Insulin bind to a specific **receptor tyrosine kinase.**
- Insulin also binds to the IGF-I receptor.
- IGF-I & IGF-II bind to all 3.
- The purple boxes are intracellular tyrosine kinase domains.



Barrett et al., 2010



IRS, insulin receptor substrate;
PI, phosphatidylinositol;
MAP, mitogen-activated protein

Holt and Hanley, 2012

- Chronic exposure to high insulin levels, obesity & excess GH → a down-regulation of insulin receptors
- Exercise & starvation → up-regulate the number of receptors

Effect of insulin

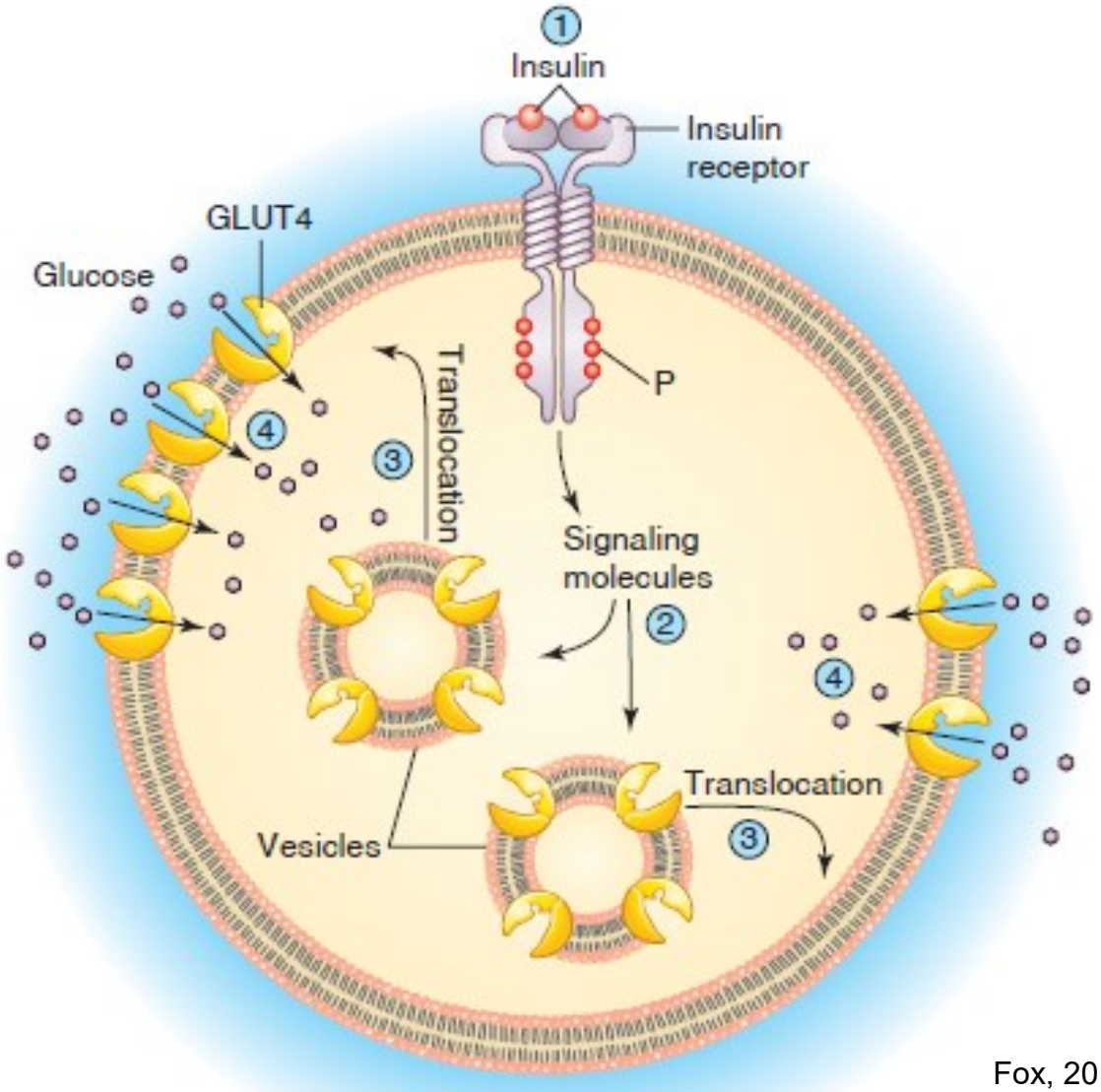
- **Hypoglycemic effect**
- **Storage of carbohydrate, protein & fat**
- Additional effects on amino acid & electrolyte transport, many enzymes, and growth

Liver
Decreased ketogenesis
Increased protein synthesis
Increased lipid synthesis
Decreased glucose output due to decreased gluconeogenesis, increased glycogen synthesis, and increased glycolysis
General
Increased cell growth

Adipose tissue
Increased glucose entry
Increased fatty acid synthesis
Increased glycerol phosphate synthesis
Increased triglyceride deposition
Activation of lipoprotein lipase
Inhibition of hormone-sensitive lipase
Increased K^+ uptake
Muscle
Increased glucose entry
Increased glycogen synthesis
Increased amino acid uptake
Increased protein synthesis in ribosomes
Decreased protein catabolism
Decreased release of gluconeogenic amino acids
Increased ketone uptake
Increased K^+ uptake

Barrett et al., 2010

Insulin stimulates uptake of blood glucose.



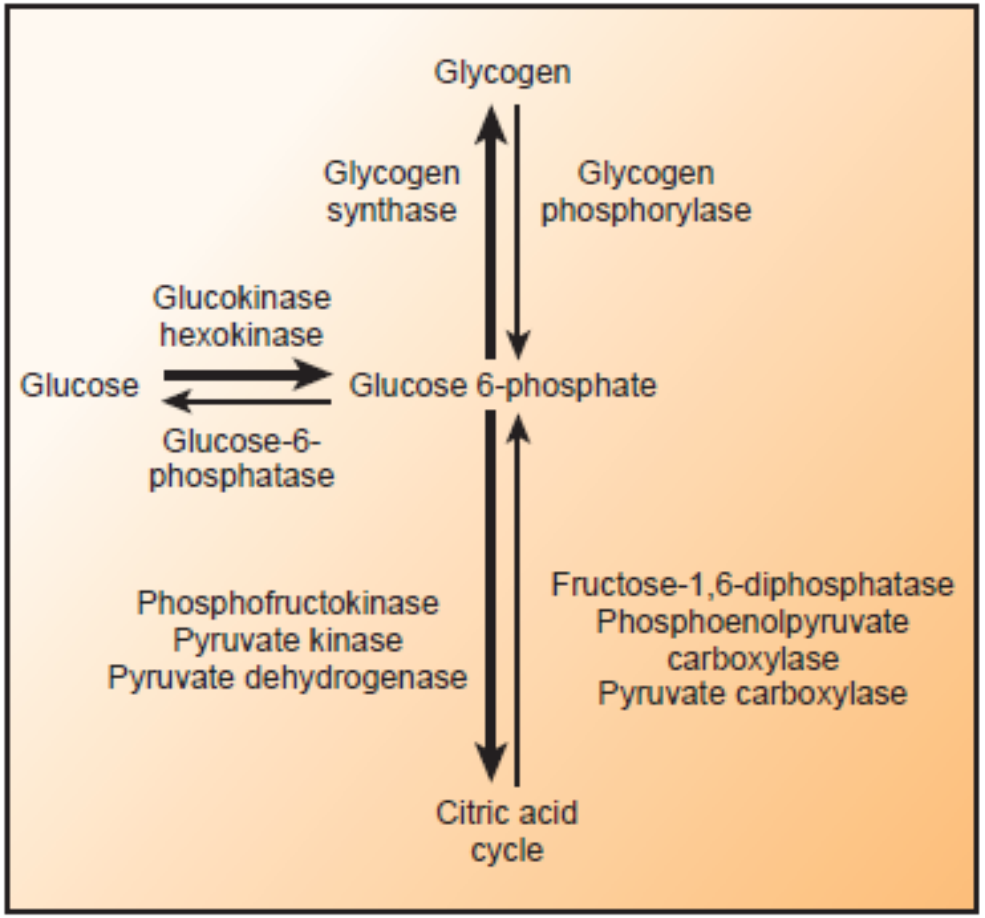
Fox, 2011

Insulin stimulates glycogen synthesis.

→ ⇧ **glycogen storage**
- ⇧ **glycogen synthase**
- ⇩ **glycogen phosphorylase**

→ ⇧ **glycolysis**
- ⇧ glucose uptake
- ⇧ glucokinase (liver)/hexokinase (muscle), phosphofructokinase, pyruvate kinase & pyruvate dehydrogenase

→ ⇩ **gluconeogenesis**
- ⇩ fructose-1,6-diphosphatase, phosphoenolpyruvate carboxylase & pyruvate carboxylase



Rhoades and Bell, 2013

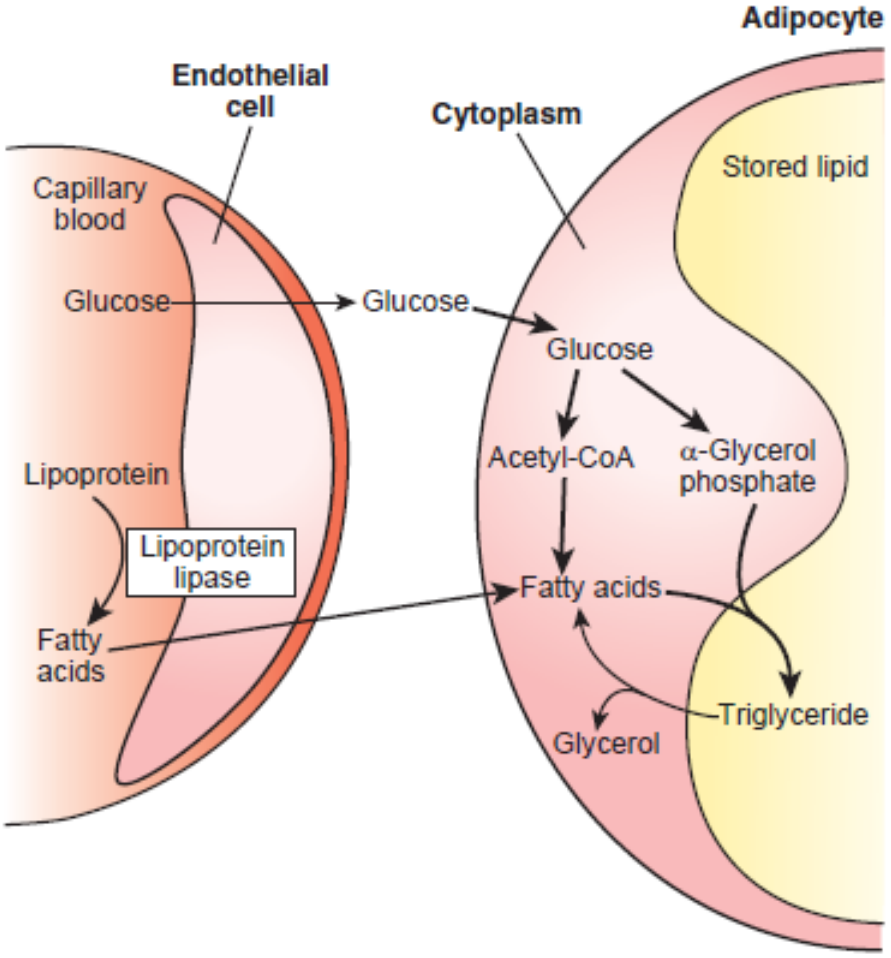
Bold arrows: processes stimulated by insulin
Light arrows: processes inhibited by insulin

Effects of insulin on lipid metabolism in adipocytes

In adipose tissue & liver tissue:

⇧ lipogenesis & ⇩ lipolysis

- ⇧ formation of α-glycerol phosphate & fatty acids (for triglyceride formation)
- ⇧ fatty acid synthase → ⇧ fatty acid synthesis
- ⇩ hormone-sensitive lipase → ⇩ breakdown of triglycerides
- ⇧ activity of lipoprotein lipase → ⇧ uptake of fatty acids from the blood into adipose tissue

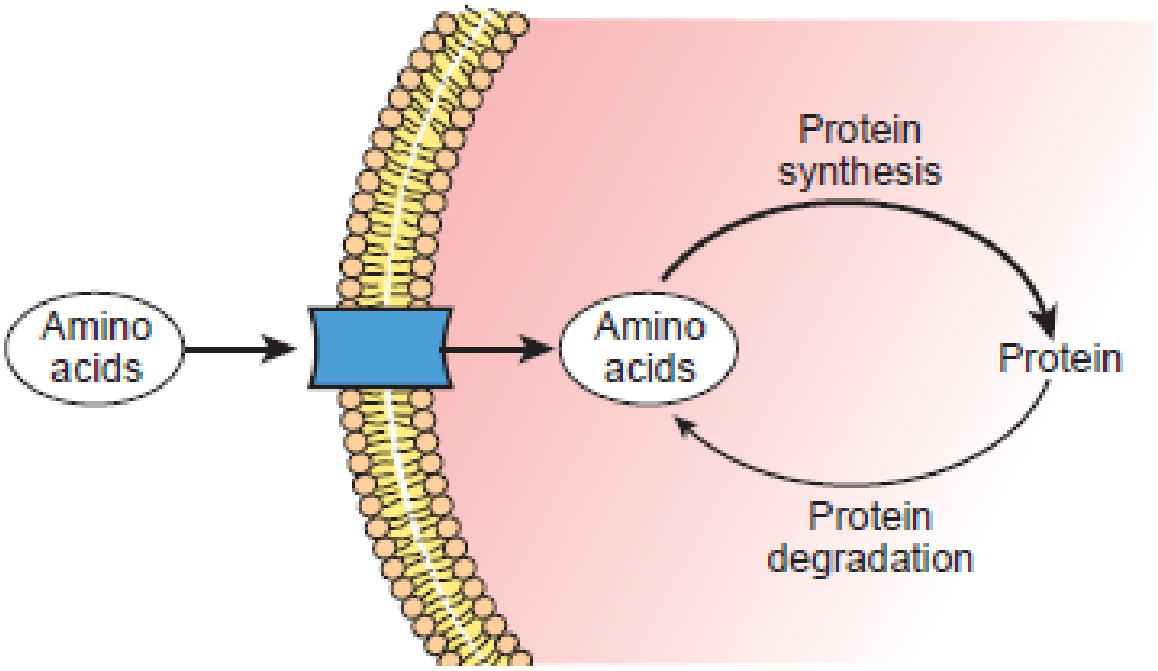


Rhoades and Bell, 2013

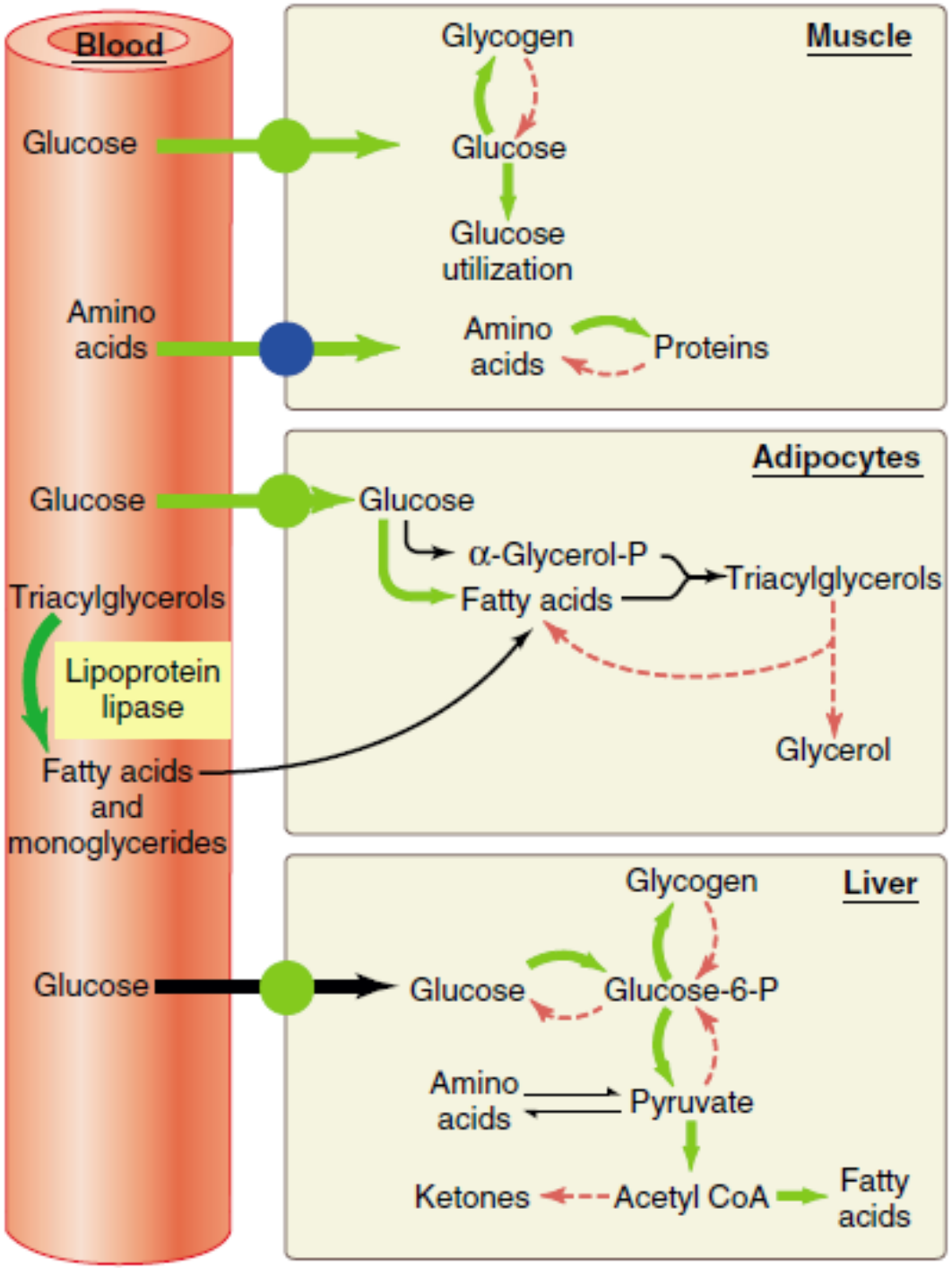
Effects of insulin on protein synthesis/degradation

⇧ protein accumulation in liver, adipose tissue & muscle in 3 specific ways.

1) ⇧ amino acid uptake
2) ⇧ activity of several factors involved in protein synthesis
3) ⇩ protein degradation (⇩ lysosome activity & possibly other mechanisms)



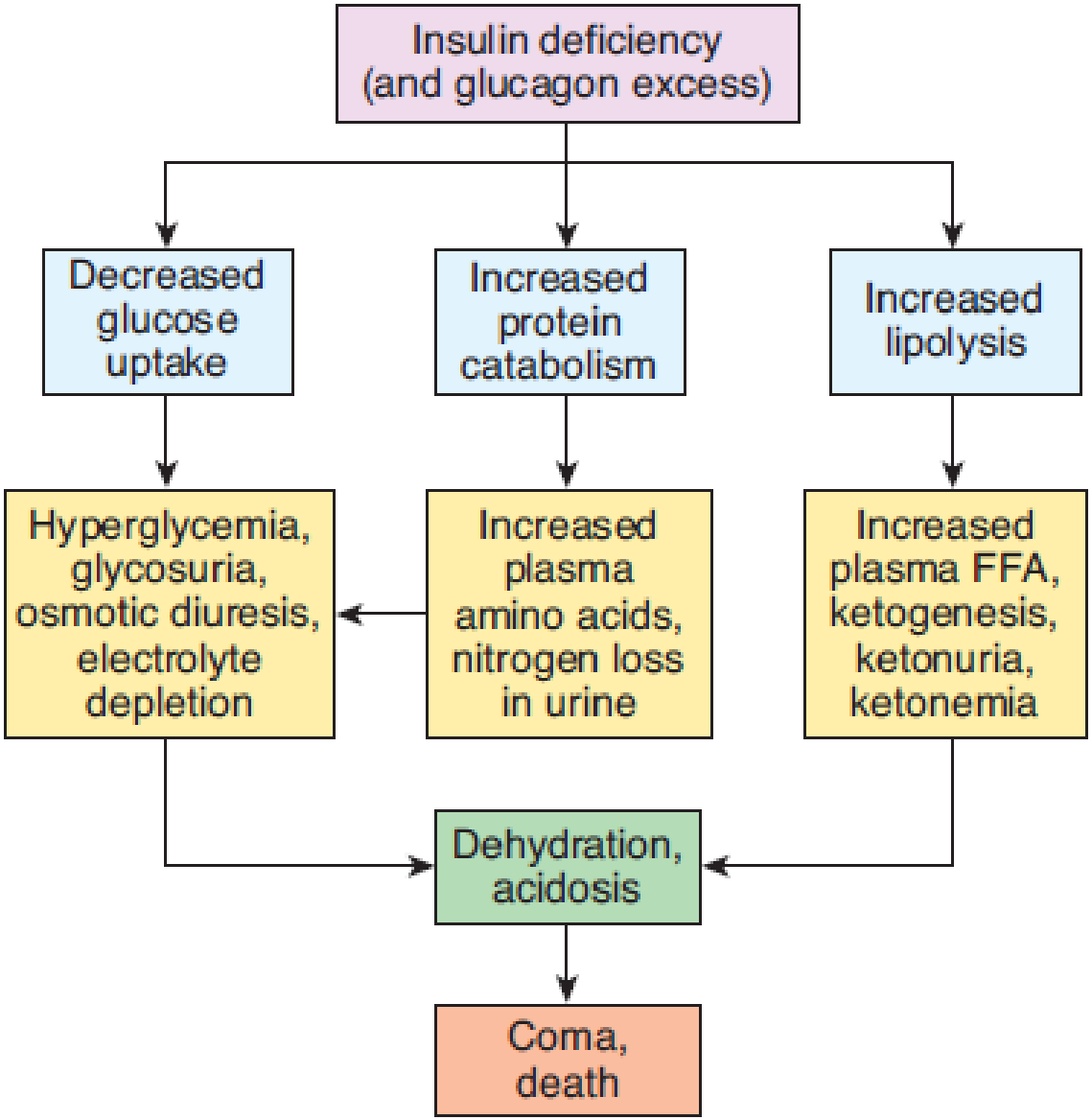
Rhoades and Bell, 2013



- green arrow: a process stimulated by insulin
- a dashed red arrow: inhibition by insulin.
- black arrows: processes that are not *directly* affected by insulin but are enhanced in the presence of increased insulin

Vander et al., 2001

Effects of insulin deficiency

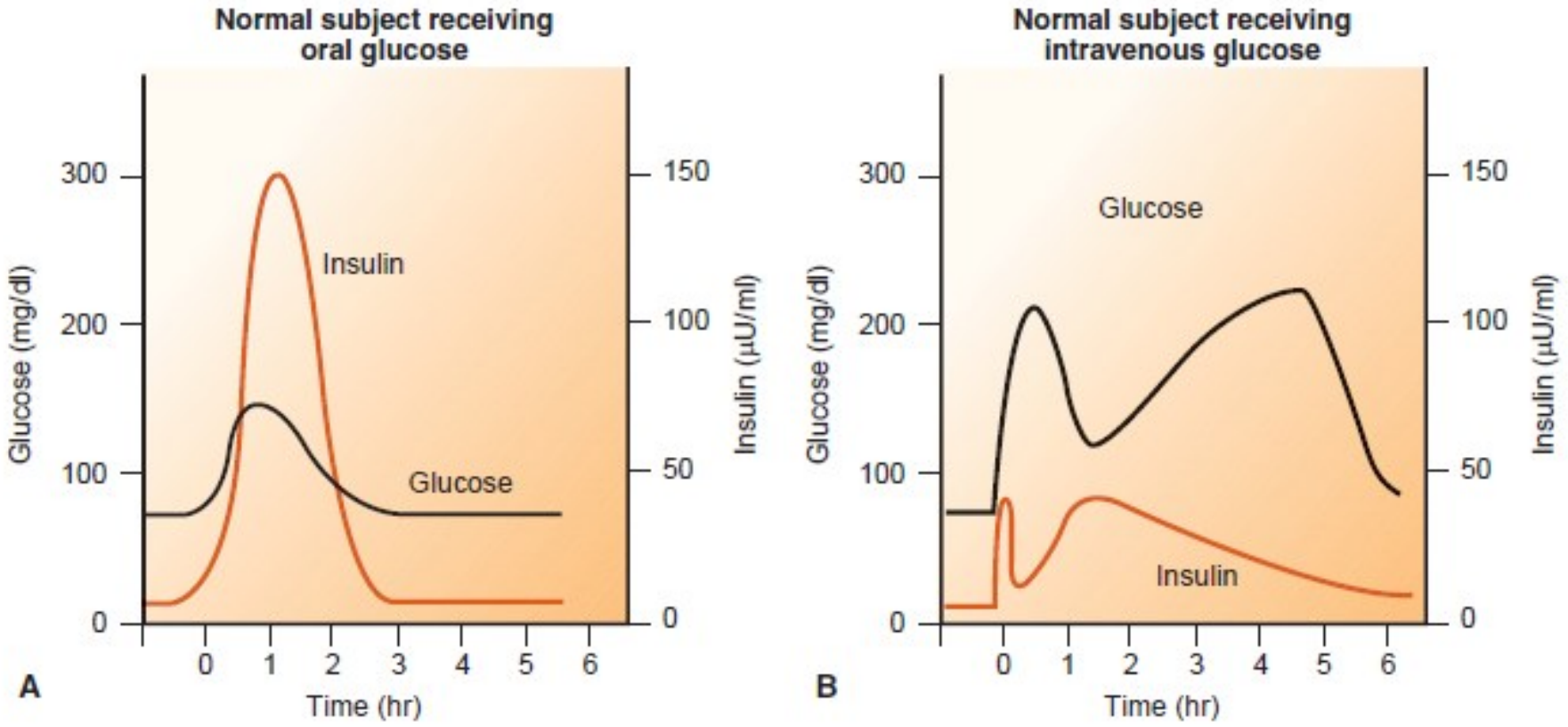


Insulin deficiency

- **Diabetes Mellitus (DM)**: a systemic disease, the body does not produce and/or use insulin, and characterized most notably by **hyperglycemia**
- **Type 1 (insulin-dependent) DM** (~ 10% of all cases of DM): a lack of insulin secretion because of erroneous autoimmune attack against the pancreatic β cells. Usually present in children, require exogenous insulin for survival.
- **Type 2 (non-insulin-dependent) DM**: insulin secretion may be normal or even increased, but insulin's target cells are less sensitive than normal. Usually present in overweight or obese adults.
- **Symptoms**
  - Fasting state (>126 mg/dL)
  - ⇧ plasma osmolarity, polydipsia (thirst)
  - Urinary loss of glucose, polyuria (loss of water)
  - Polyphagia (hunger), weight loss
  - ⇧ lipolysis & proteolysis
  - ⇧ ketone bodies in the blood (diabetic ketoacidosis)

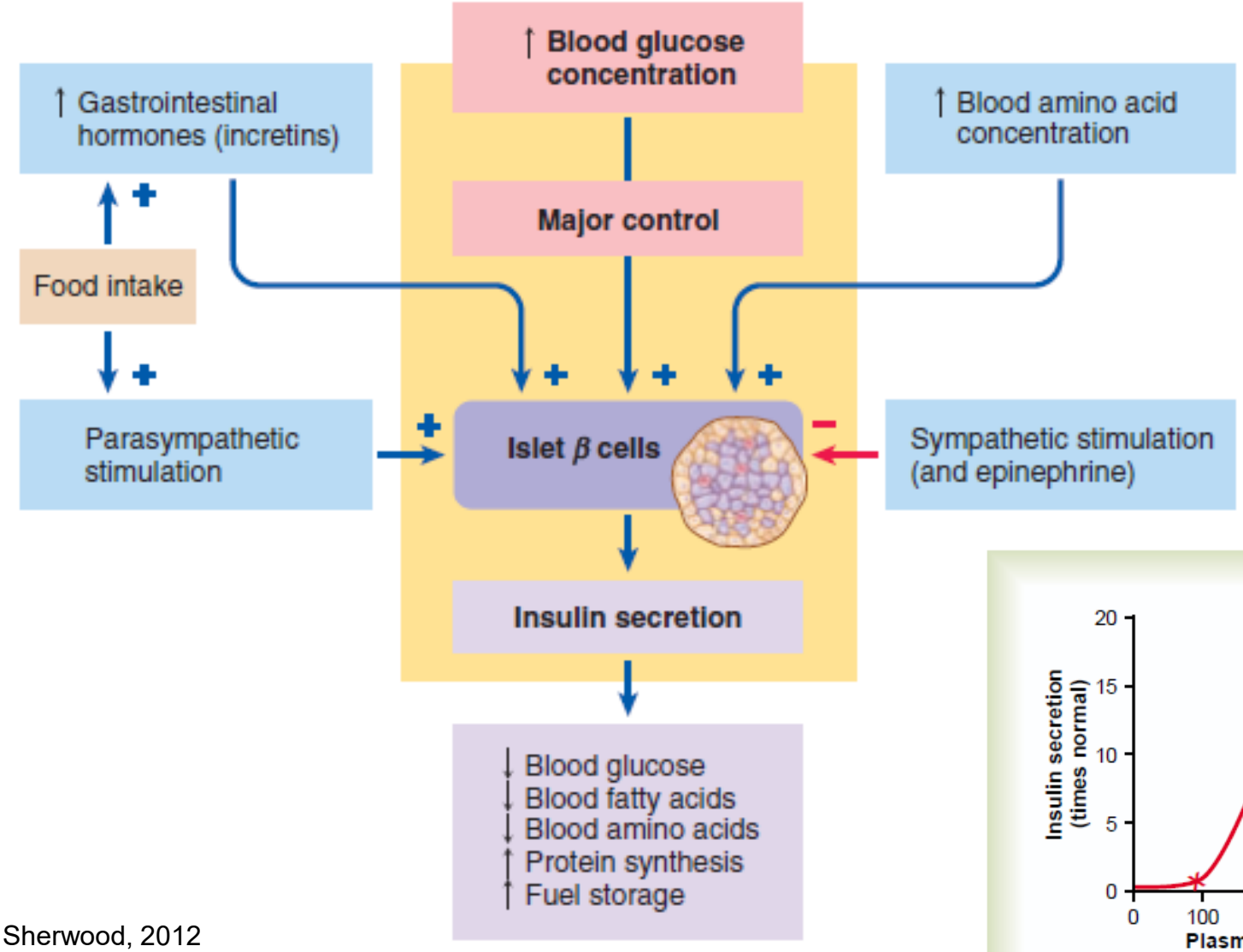
Plasma glucose & insulin following an oral or i.v. glucose challenge

incretin effect

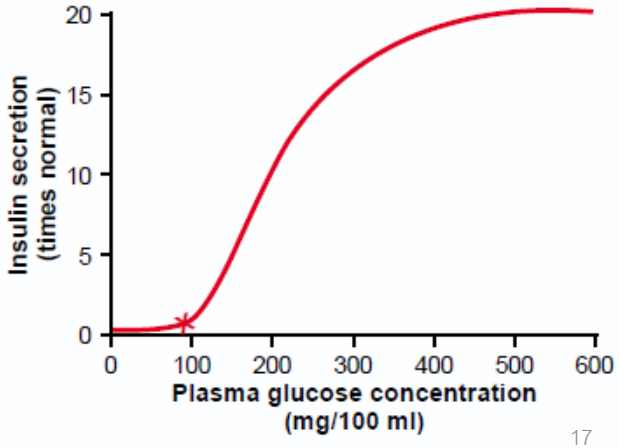


Rhoades and Bell, 2013

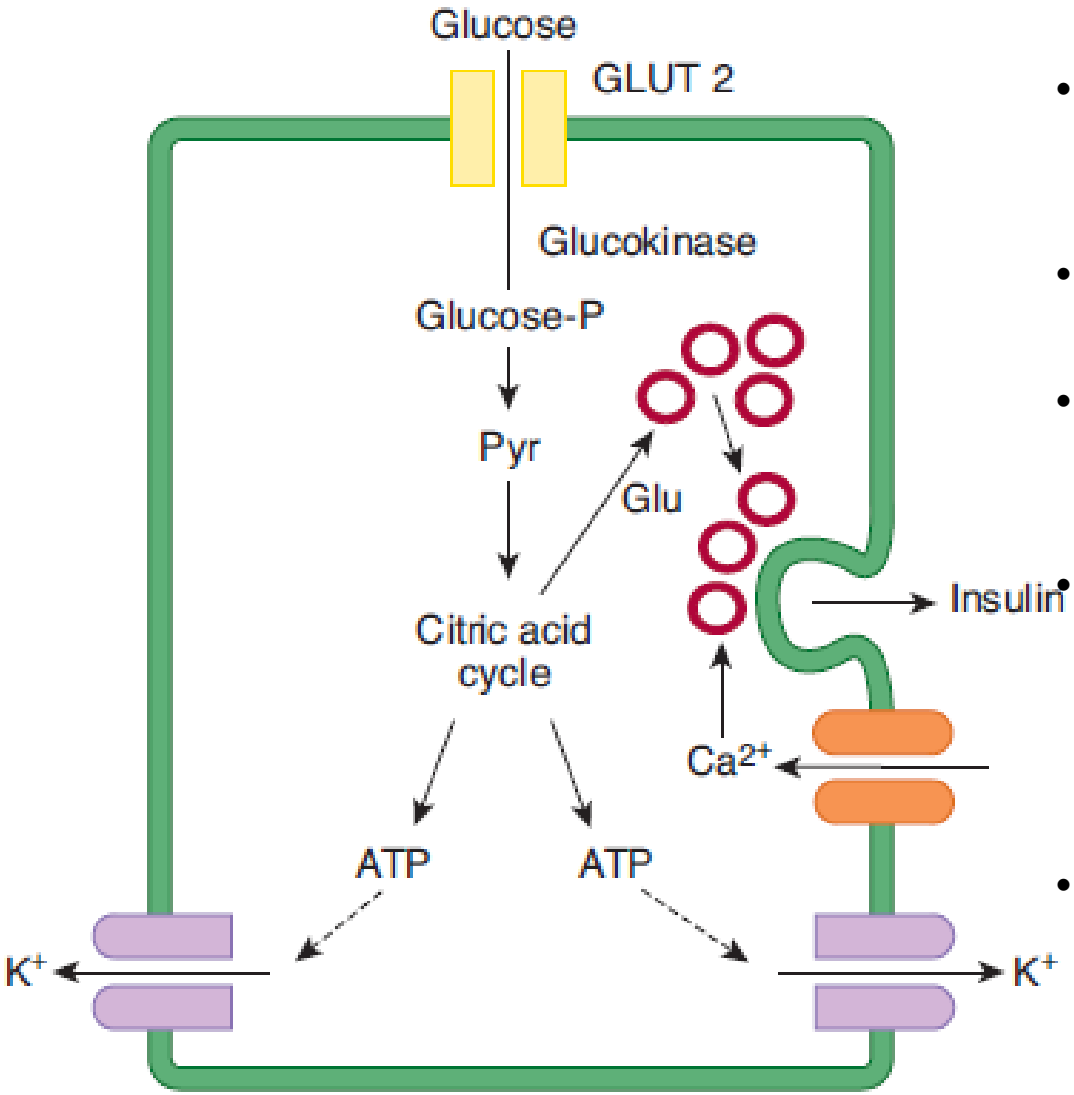
Regulation of insulin secretion



Sherwood, 2012

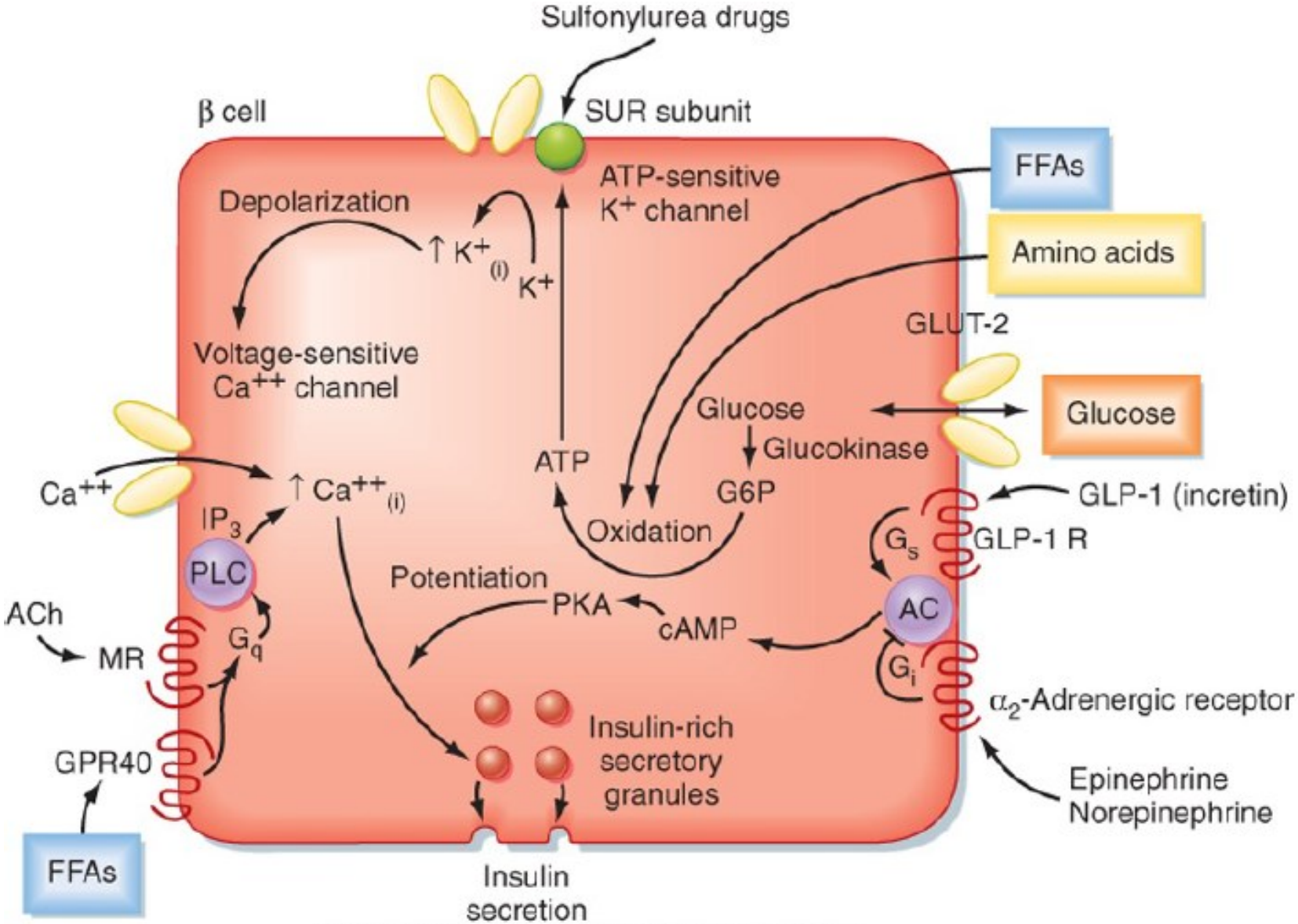


Regulation of insulin secretion



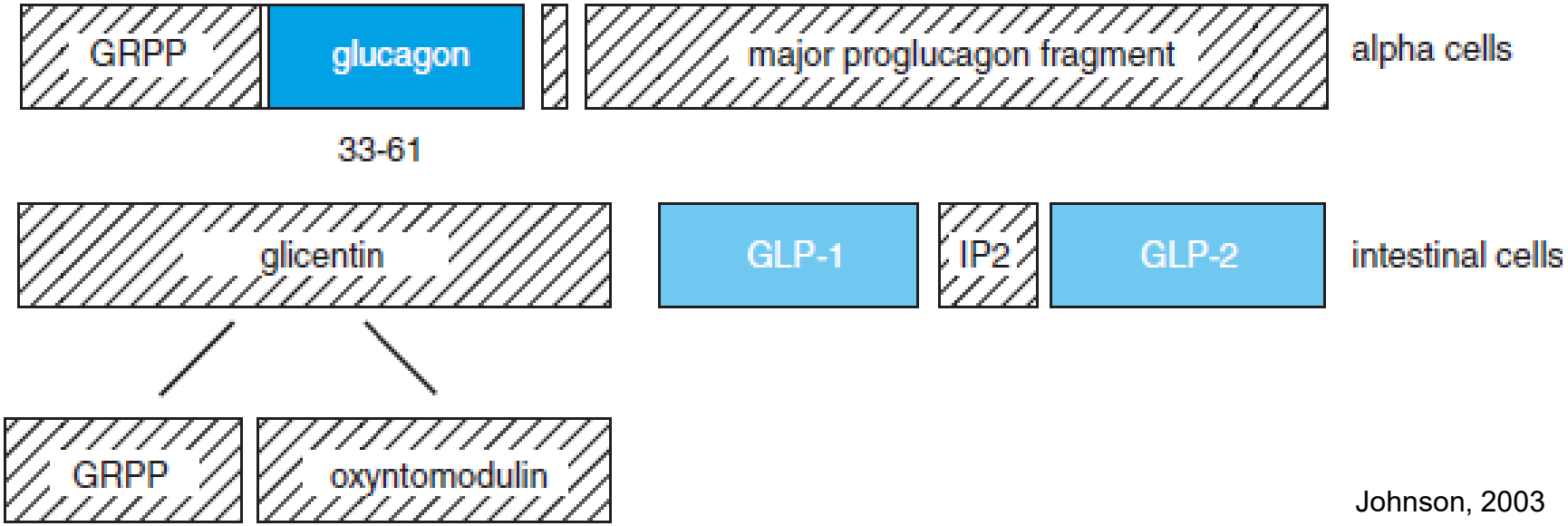
- Glucose is phosphorylated to G6P by **glucokinase (the glucose sensor)**
- Glucose is metabolized to pyruvate (Pyr) in the cytoplasm.
- Pyr enters the mitochondria and is metabolized via the citric acid cycle.
- ATP formed by oxidative phosphorylation → ⇩ ATP-sensitive K^+ channels → ⇩ K^+ efflux → B cell depolarized → ⇧ Ca^{2+} influx → ⇧ release of insulin by exocytosis
- Glutamate (Glu) is also formed & this primes secretory granules, preparing them for exocytosis.

Barrett et al., 2010



Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.
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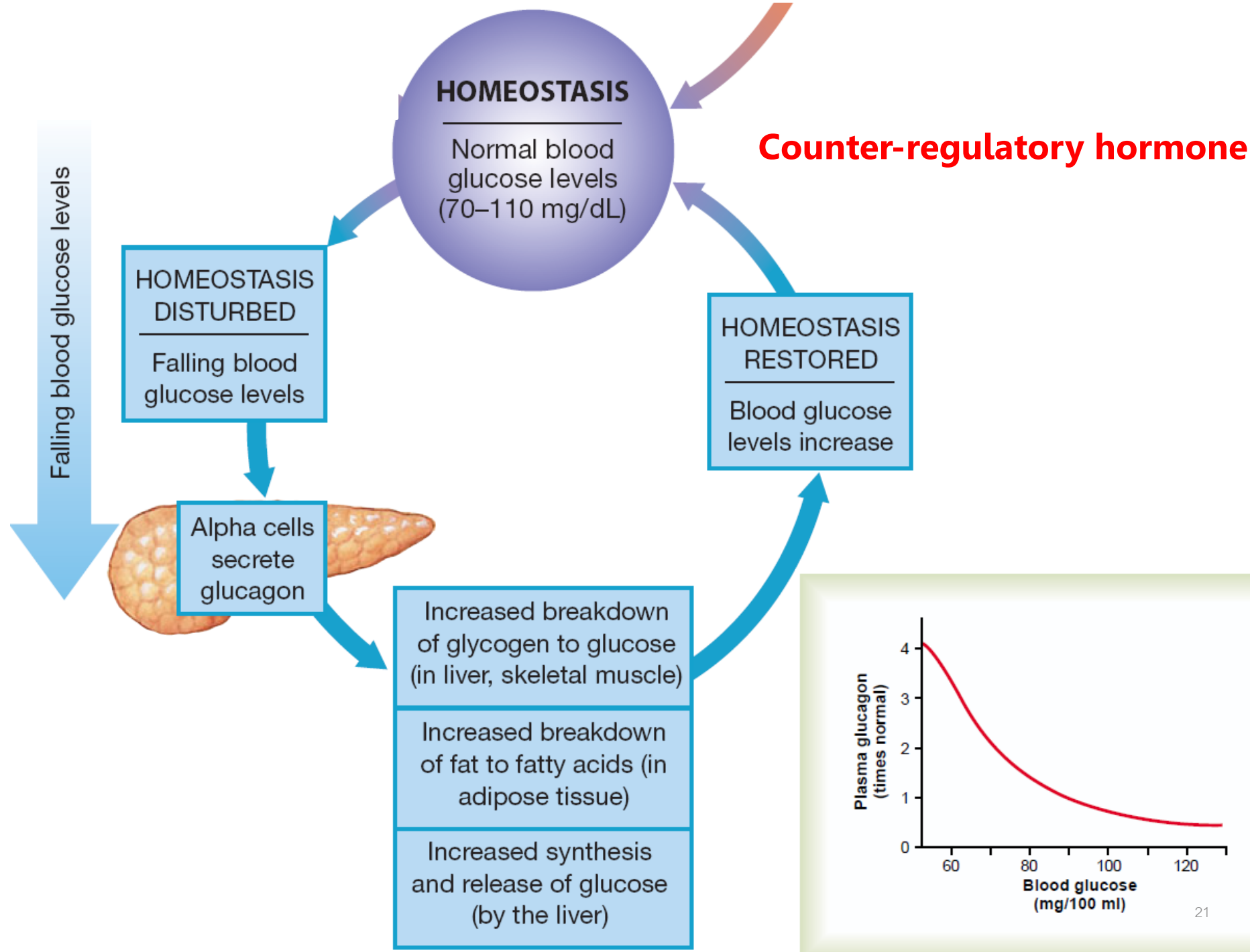
Glucagon



Johnson, 2003

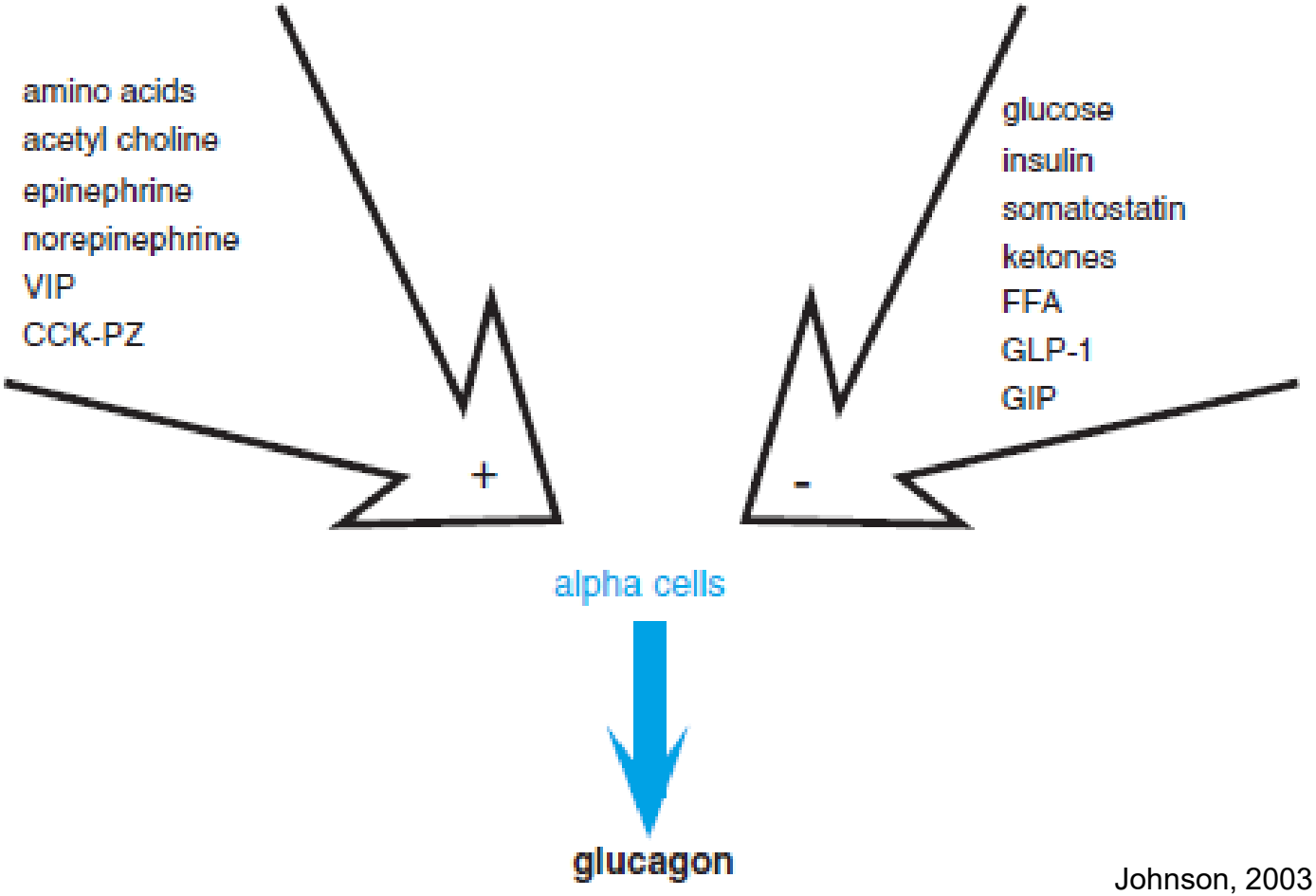
- **Target tissues: liver**, cardiac & skeletal muscle, and adipose tissue
- **Functions: stimulating glycogenolysis, gluconeogenesis & ketogenesis**

- Cell-specific post-translational processing of preproglucagon
- GRPP, glicentin related pancreatic peptide; GLP-1, glucagon-like peptide-1; GLP-2, glucagon-like peptide-2; IP2, intervening peptide 2



Counter-regulatory hormone

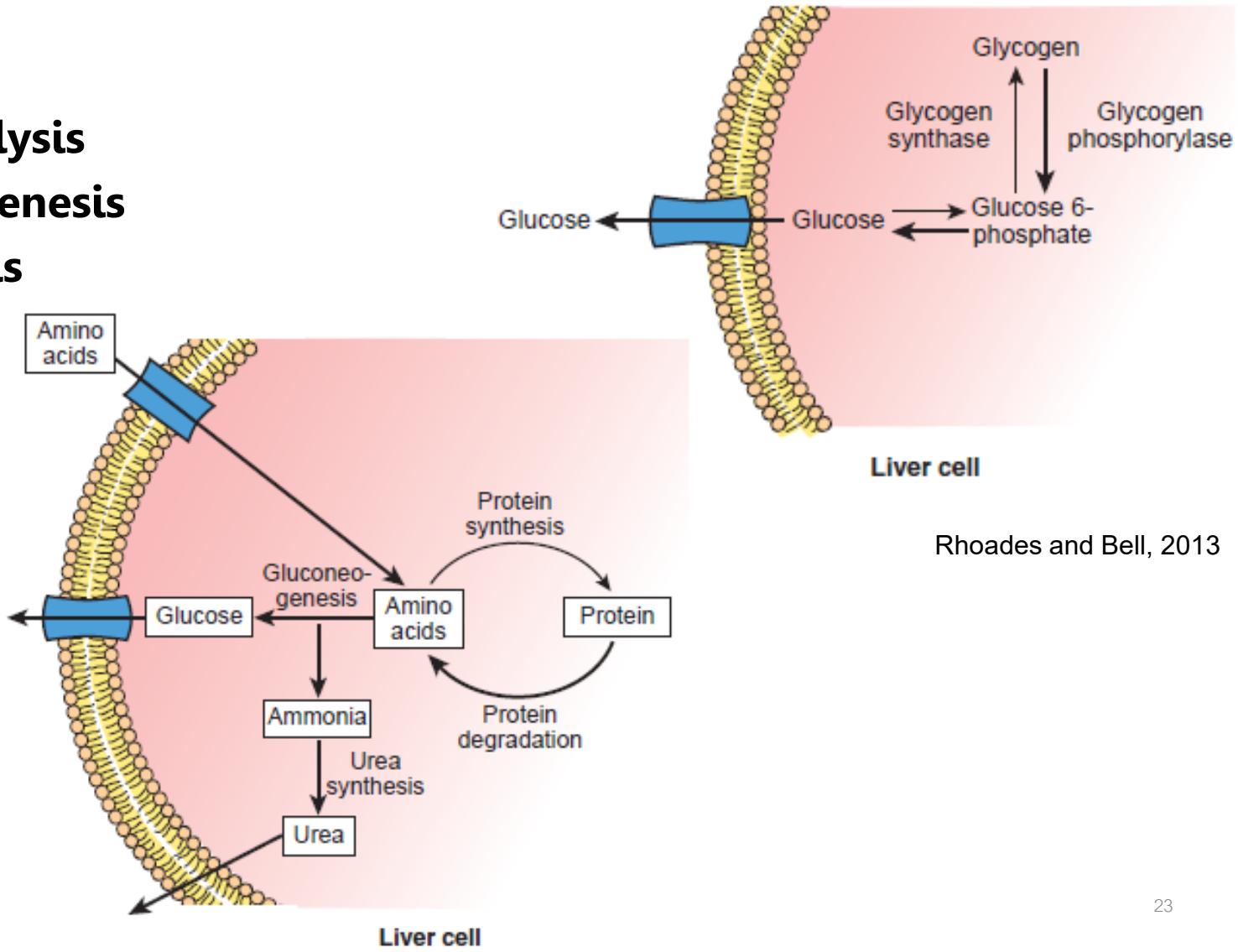
Stimulatory & inhibitory signals for glucagon secretion



Johnson, 2003

Glucagon influences the metabolism of carbohydrates, lipids & proteins primarily in the liver

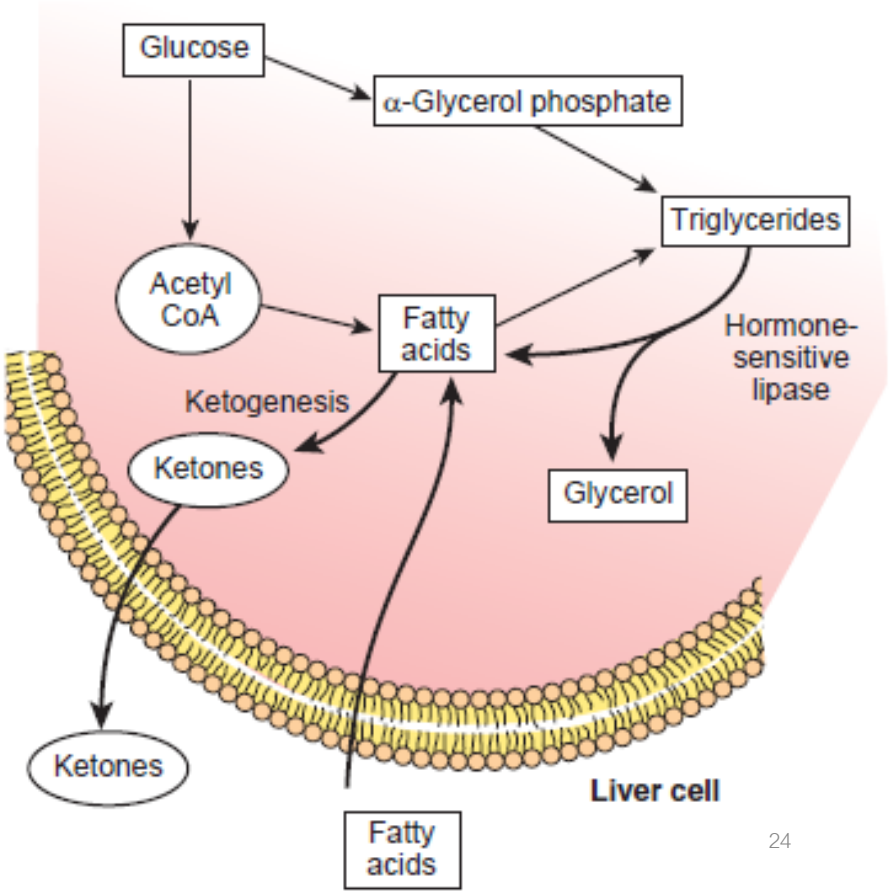
- **Glycogenolysis**
- **Gluconeogenesis**
- **Ureagenesis**



Rhoades and Bell, 2013

- **Lipolysis**
- **Ketogenesis**

- Ketones: an important source of fuel for muscle & heart cells during times of starvation, sparing blood glucose for other tissues that are obligated glucose users, e.g., CNS.
- During prolonged starvation, the brain adapts its metabolism to use ketones as a fuel source, lessening the overall need for hepatic glucose production.



Rhoades and Bell, 2013



"I think it's called a scale,
but mom calls it a @#$&% liar!"